Sheth L.U.J. And Sir M.V. College

Aim: Sensor Mote and Base Station Communication
Study Develop a simple communication scenario between multiple sensor motes and a base station to study data aggregation and sink node operation.

Code:

SensorNode.ned:-

```
package sensornetwork;

simple SensorNode
{
    parameters:
        double sendInterval @unit(s);
    gates:
        output out;
}
```

BaseStation.ned:-

```
package sensornetwork;

simple BaseStation
{
    gates:
        input in[];
}
```

SensorNetwork.ned:-

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```
package sensornetwork;

network SensorNetwork
{
    submodules:
        sensor[5]: SensorNode {
            parameters:
                sendInterval = 2s;
        }

        sink: BaseStation;

    connections:
        sensor[0].out --> sink.in++;
        sensor[1].out --> sink.in++;
        sensor[2].out --> sink.in++;
        sensor[3].out --> sink.in++;
        sensor[4].out --> sink.in++;
}
```

SensorNode.cc:-

```
#include <omnetpp.h>

class SensorNode : public cSimpleModule
{
  private:
    cMessage *timer;

  protected:
    virtual void initialize()
    {
        timer = new cMessage("timer");
        scheduleAt(simTime(), timer);
    }

    virtual void handleMessage(cMessage *msg)
    {
        if(msg == timer)
        {
```

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```
            cMessage *pkt = new cMessage("SensorData");
            send(pkt, "out");

            scheduleAt(simTime() + par("sendInterval").doubleValue(), timer);
        }
    }

    virtual ~SensorNode()
    {
        cancelAndDelete(timer);
    }
};

Define_Module(SensorNode);
```

BaseStation.cc

```
#include <omnetpp.h>

class BaseStation : public cSimpleModule
{
  private:
    int count;

  protected:
    virtual void initialize()
    {
        count = 0;
    }

    virtual void handleMessage(cMessage *msg)
    {
        count++;

        EV << "Packet received. Total = "
           << count << "\n";

        delete msg;
    }
```

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```
    virtual void finish()
    {
        recordScalar("TotalPackets", count);
    }
};

Define_Module(BaseStation);
```

omnetpp.ini

```
[General]
network = SensorNetwork

sim-time-limit = 50s

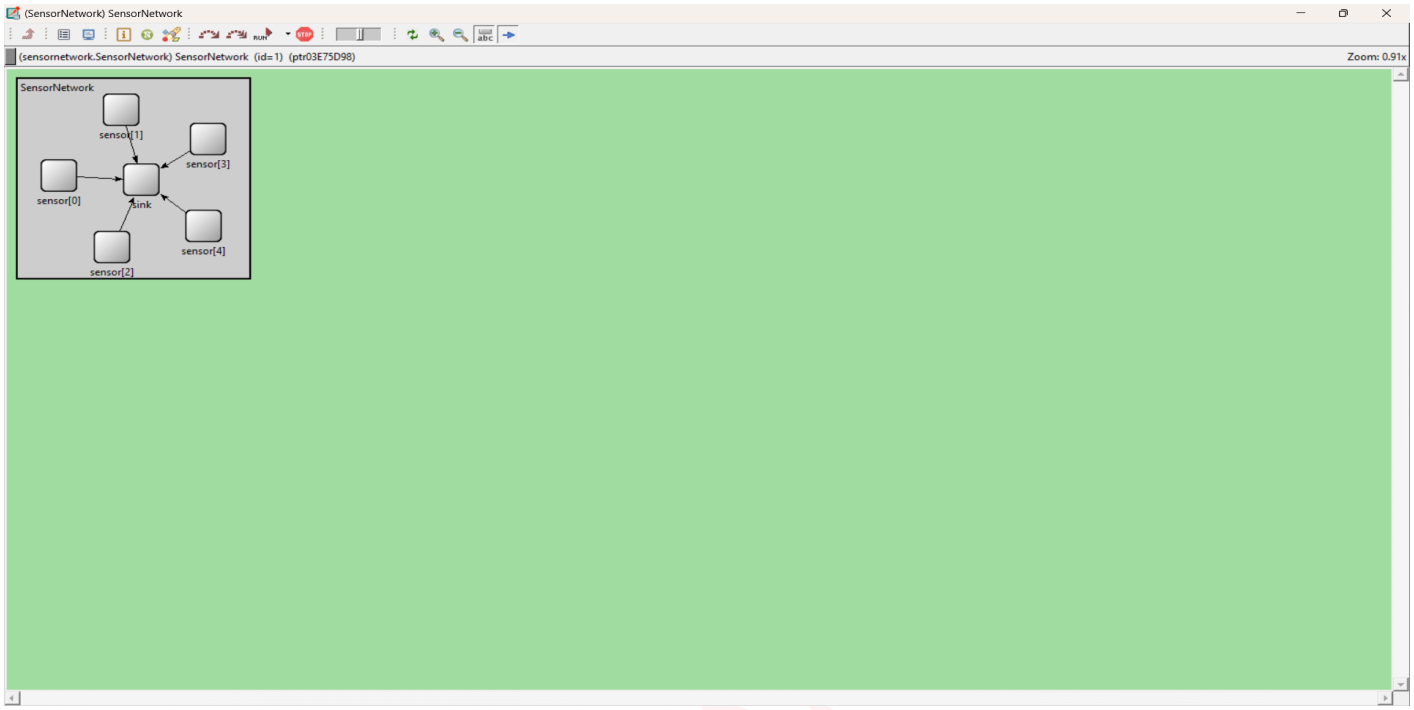
*.sensor[*].sendInterval = 2s
```

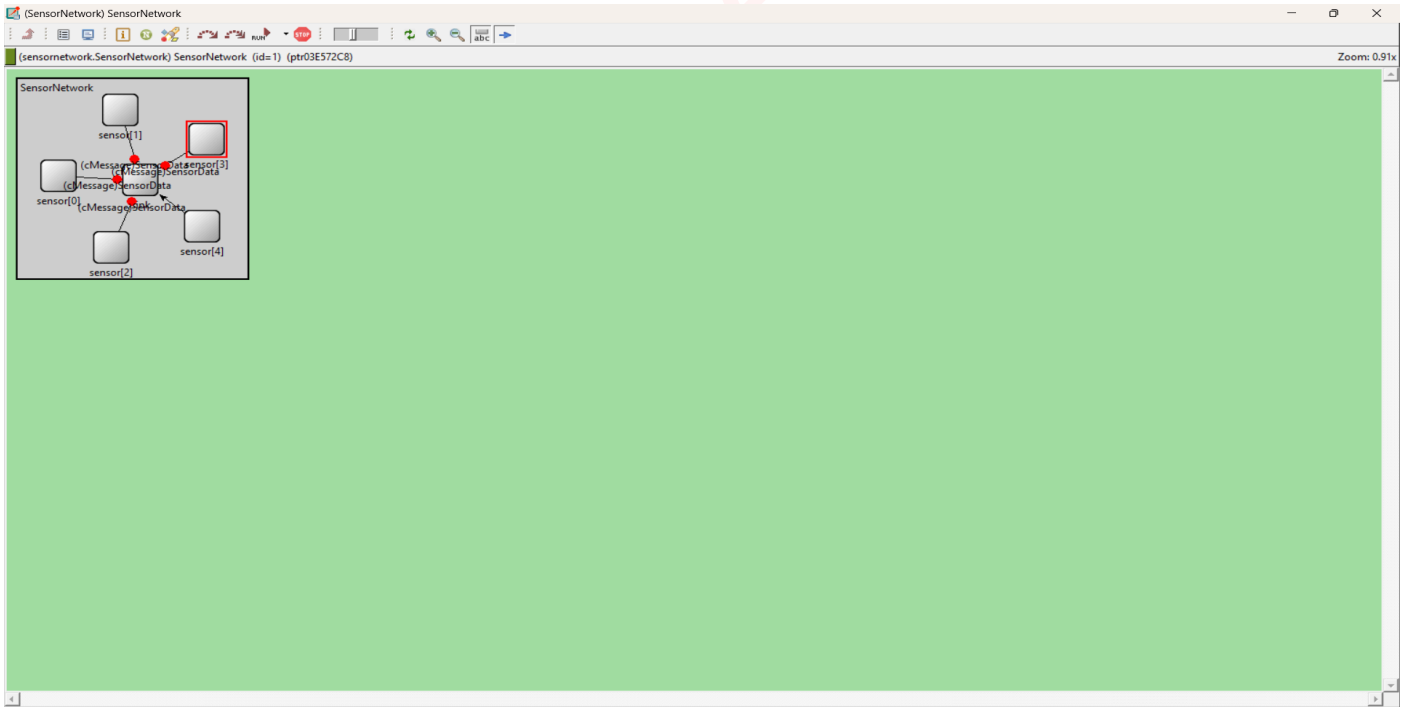
OUTPUT:
NETWORK TOPOLOGY:

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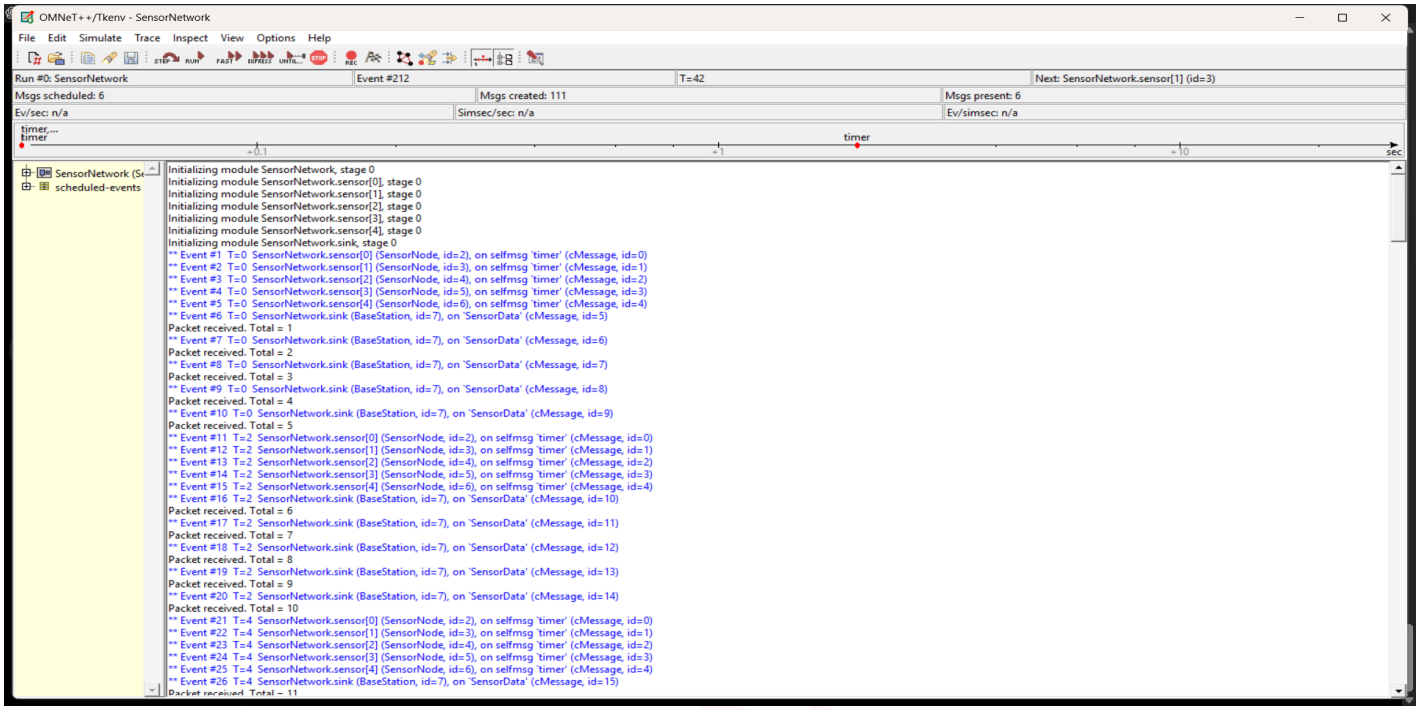
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Result

The sensor motes successfully transmitted data to the base station, and the sink node aggregated all received packets during the simulation.

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